

MAHESH BOOPATHI S

AI Engineer | Gen AI Specialist

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PROFESSIONAL SUMMARY

- AI Engineer with 2+ years of experience building production-grade Agentic AI systems, RAG pipelines, and LLM fine-tuning workflows (LoRA, PEFT, QLoRA) on AWS SageMaker and Azure Machine Learning, with strong expertise in Python.
- Deployed a real-time, low-latency AI voice sales agent on the OpenAI Realtime API with Pinecone-backed RAG, achieving 90%+ context-aware retrieval accuracy in live production use.
- Designed MCP-integrated multi-agent workflows that cut manual processing time by 60%, and delivered a cost-efficient LLM fine-tuning platform that improved response accuracy by 40% through RAG layering.

TECHNICAL SKILLS

Generative AI: LLMs, RAG, Prompt Engineering, Embeddings, LangChain

Agentic AI & MCP: AI Agents, Agentic Workflows, Multi-Agent Systems, MCP Server Development, A2A Protocol

LLM Fine-Tuning: LoRA, PEFT, QLoRA, Instruction Tuning, Dataset Preparation

Speech & Voice AI: ASR, Whisper, OpenAI Realtime API

Languages: Python, JavaScript, SQL

Backend & Messaging: FastAPI, Kafka, Redis

Deep Learning: PyTorch, Hugging Face Transformers

Cloud & AI Platforms: AWS SageMaker, Azure Machine Learning, Azure AI Foundry, Azure AI Services

Vector Databases: Pinecone, FAISS

MLOps & Tools: Docker, Git, CI/CD, MLflow

PROFESSIONAL EXPERIENCE

AI Engineer — SetNext.ai | Bengaluru

Apr 2025 – Aug 2026

Agentic AI Workflow Automation System

- Architected a multi-agent system executing complex enterprise workflows end to end, cutting manual processing time by 60%.
- Implemented MCP (Model Context Protocol) servers connecting AI agents to enterprise tools and data sources.
- Applied Pinecone-based RAG with embeddings to deliver context-aware responses at 90%+ query accuracy.
- Unified Azure AI Foundry, Azure AI Services, and Azure Machine Learning for model integration and inference.
- Enabled real-time KPI monitoring to track agent execution and operational performance.

Tech Stack: Python, MCP, Pinecone, Azure AI Foundry, Azure AI Services, Azure Machine Learning

AI-Powered Real-Time Voice Sales Agent

- Delivered a real-time voice sales agent on the OpenAI Realtime API for low-latency, multi-turn customer conversations.
- Engineered a RAG knowledge system with document chunking, embeddings, and vector indexing for grounded responses.
- Integrated Azure AI Foundry and Azure AI Services to trigger automated, tool-based sales workflows.
- Tracked conversation activity, response latency, and sales performance through real-time KPI monitoring.

Tech Stack: OpenAI Realtime API, Pinecone, Azure AI Foundry, Azure AI Services, Python

LLM Fine-Tuning-as-a-Service PlaVorm

- Launched a cloud-based platform for scalable model training, customization, and deployment across enterprise use cases.
- Fine-tuned LLMs with LoRA, PEFT, and QLoRA, enabling cost-effective model customization for enterprise needs.
- Exposed APIs for dataset validation, training orchestration, and multi-model versioning on AWS SageMaker and Azure ML.
- Augmented the platform with RAG pipelines and real-time KPI monitoring, improving response accuracy by 40%.

Tech Stack: AWS SageMaker, Azure Machine Learning, LoRA/PEFT/QLoRA, Pinecone, Python

Software Trainee Intern — Guru Tech

Sep 2024 – Feb 2025

ML-Based Crop Recommendation System

- Developed an ML-based crop recommendation system that translates soil and environmental data into data-driven guidance.
- Trained SVM and Logistic Regression classifiers to recommend suitable crops from soil and environmental features.
- Designed preprocessing and feature-selection pipelines, improving training reliability and reducing data-cleaning time.

Tech Stack: Python, SVM, Logistic Regression

CERTIFICATIONS

Data Analytics with Python — NPTEL, IIT Roorkee | 12-week course: probability, hypothesis testing, regression, and clustering

EDUCATION

Master of Computer Applications (MCA)

2022 – 2024

Muthyammal Engineering College, Namakkal, Tamil Nadu